Supplementary Activity

BY

Nat Mendoza

Problem

1. Create a class for Vegetable and Fruit with constructor, deconstructor, copy constructor, copy assignment operator with attributed and function

2. Create an array for the grocery list that contains all Jenna’s items

3. Create a function for totalSum that calculates the sum of all objects from Jenna list.

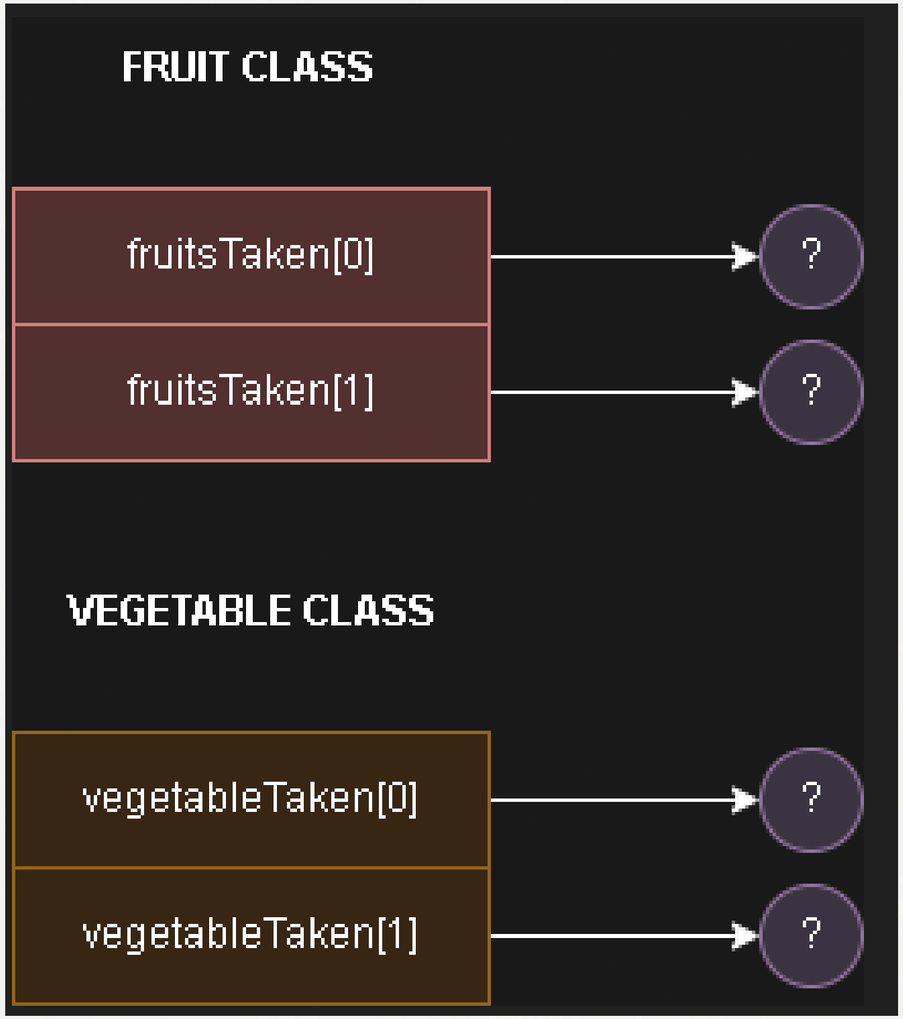
4. Delete lettuce from Jenna Receipt and de-allocate the memory assigned to that object.

Code

Visualization and Analysis

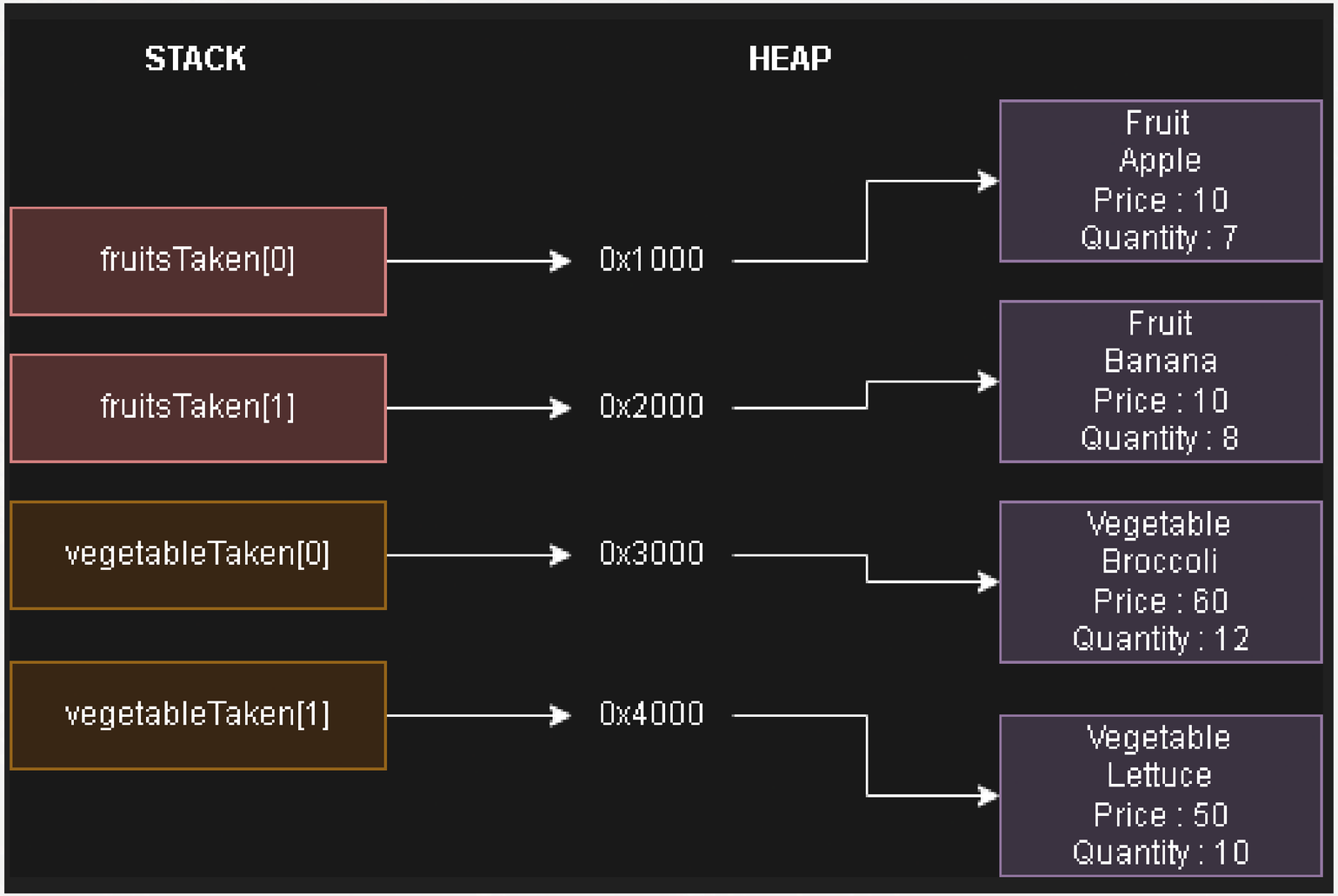
```
/* Problem 2: Creating a GroceryList Array */

Fruit* fruitsTaken[2];
Vegetable* vegetablesTaken[2];
```



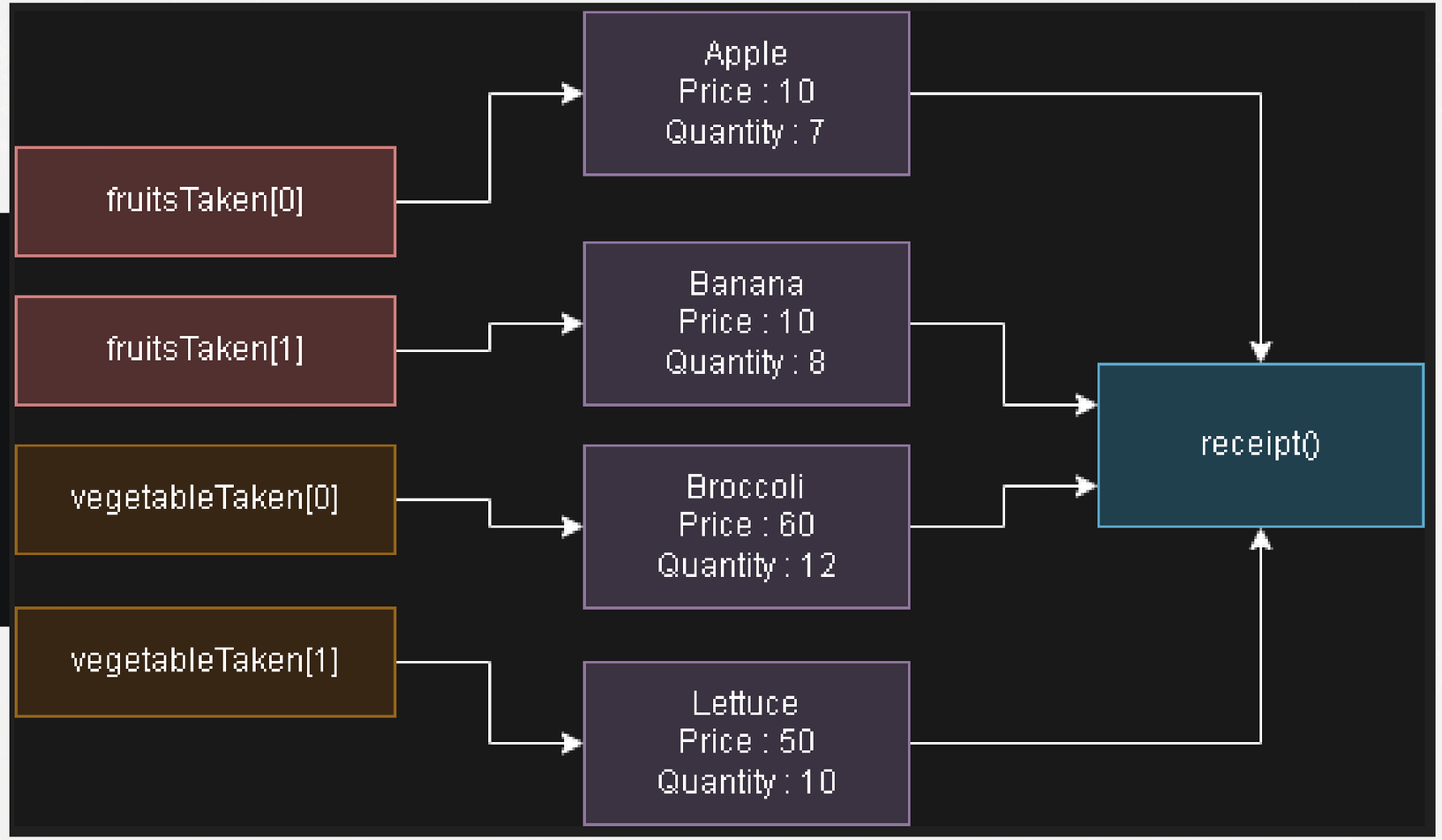
```
fruitsTaken[0] = new Fruit("Apple", 10, 7);
fruitsTaken[1] = new Fruit("Banana", 10, 8);

vegetablesTaken[0] = new Vegetable("Broccoli", 60, 12);
vegetablesTaken[1] = new Vegetable("Lettuce", 50, 10);
```



```cpp
std::cout << "Fruits:" << std::endl;
for (int i = 0; i < 2; i++) {
    fruitsTaken[i]->receipt();
}

std::cout << "Vegetables:" << std::endl;
for (int i = 0; i < 2; i++) {
    vegetablesTaken[i]->receipt();
}
```



```cpp
double totalSum(Fruit* fruits[], int fruitCount,
                Vegetable* vegetables[], int vegetableCount) {

    double total = 0;

    for (int i = 0; i < fruitCount; i++) {
        if (fruits[i] != nullptr) {
            total += fruits[i]->totalSum();
        }
    }

    for (int i = 0; i < vegetableCount; i++) {
        if (vegetables[i] != nullptr) {
            total += vegetables[i]->totalSum();
        }
    }

    return total;
}
```

```
============================================================
                  JENNA'S GROCERY RECEIPT
============================================================
Type            Item            Price     Quantity  Total
------------------------------------------------------------
Fruits:
Fruit           Apple           10          7           70
Fruit           Banana          10          8           80
Vegetables:
Vegetables      Broccoli        60          12          720
Vegetables      Lettuce         50          10          500

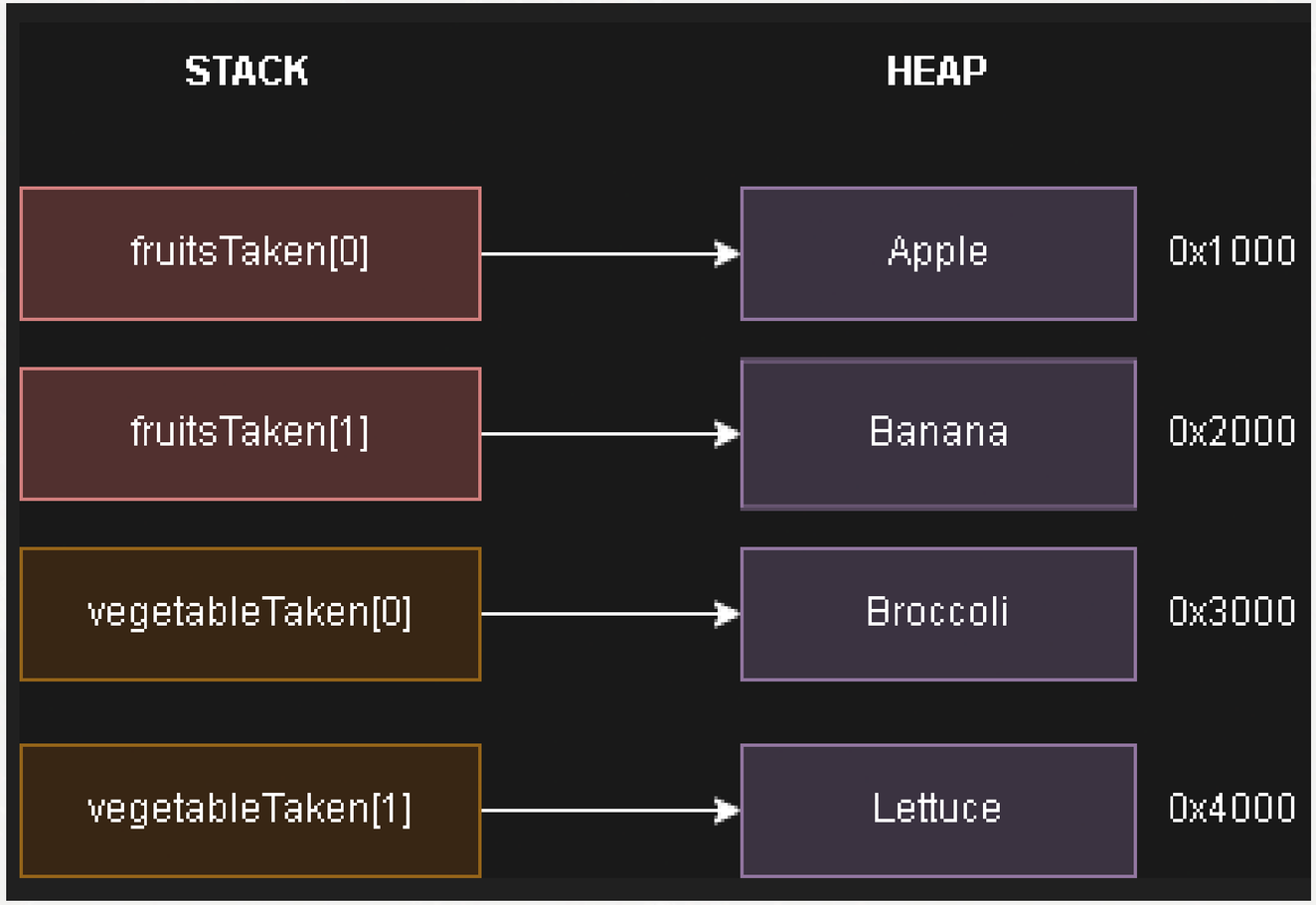
Total Amount in Pesos: 1370
```

```cpp
std::cout << "\n========== POINTER LIST BEFORE DELETION ==========\n";

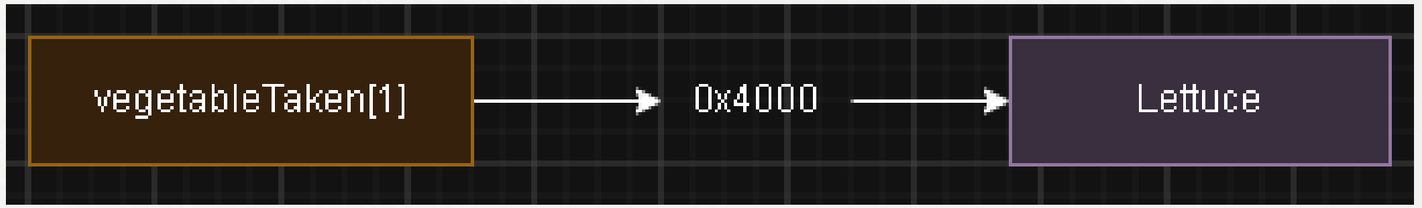
std::cout << "Apple Pointer    : " << fruitsTaken[0] << std::endl;
std::cout << "Banana Pointer   : " << fruitsTaken[1] << std::endl;
std::cout << "Broccoli Pointer : " << vegetablesTaken[0] << std::endl;
std::cout << "Lettuce Pointer  : " << vegetablesTaken[1] << std::endl;

std::cout << "================================================\n";
```

```
================ MEMORY STATUS ========
Apple Pointer       : 0x23da7847430
Banana Pointer      : 0x23da78473f0
Broccoli Pointer    : 0x23da7847070
Lettuce Pointer     : 0x23da7846fb0
=======================================
```



BEFORE



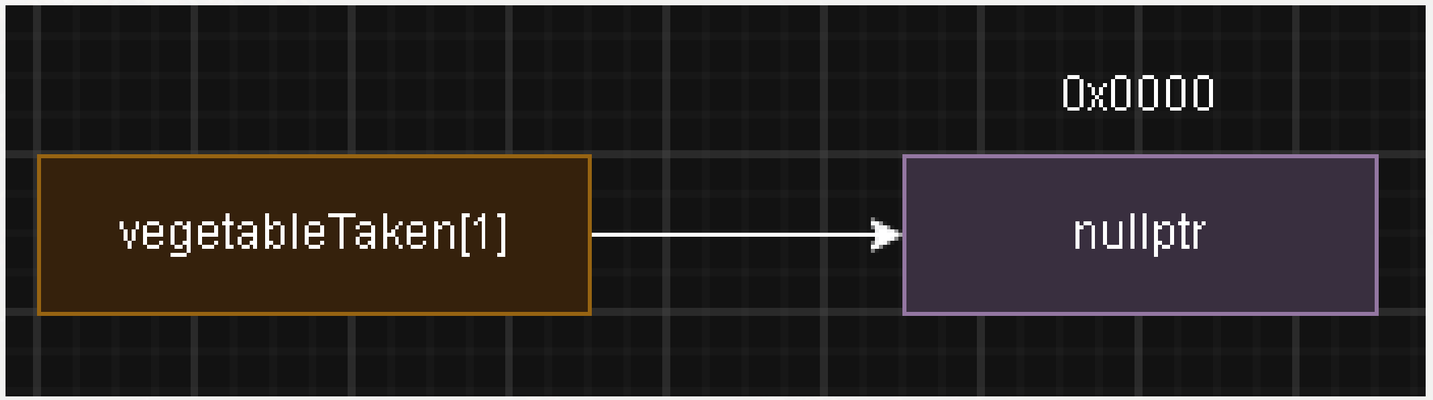

```
delete vegetablesTaken[1];
```

AFTER



MEMORY RELEASED

```
vegetablesTaken[1] = nullptr;
```

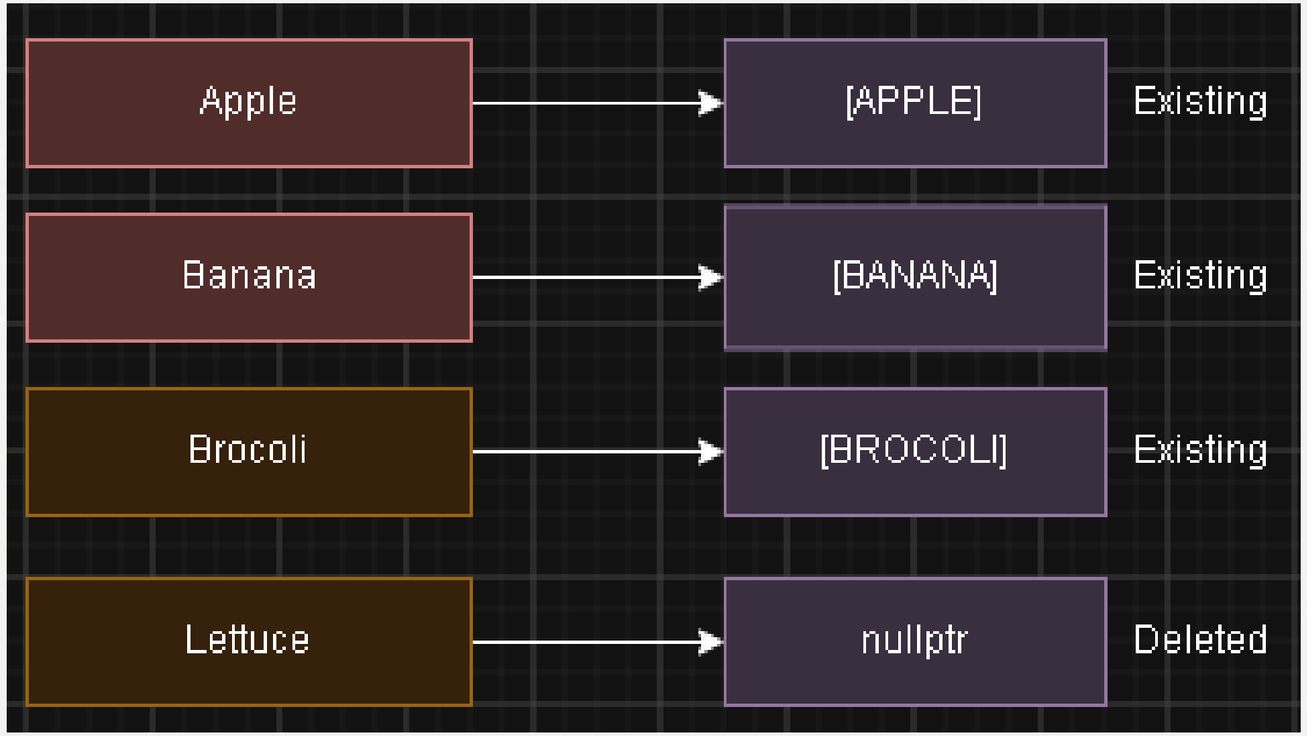


```
Deleting Lettuce...
Vegetable Destructor Called: Lettuce

Memory successfully deallocated.

========== POINTER LIST AFTER DELETION ==========
Apple Pointer    : 0x19b0ea971f0
Banana Pointer   : 0x19b0ea97470
Broccoli Pointer : 0x19b0ea976f0
Lettuce Pointer  : 0
================================================

Lettuce successfully deleted.
Lettuce pointer is now nullptr.
```



```
==========================================================
              UPDATED GROCERY RECEIPT
==========================================================
Fruits:
Fruit           Apple           10          7           70
Fruit           Banana          10          8           80

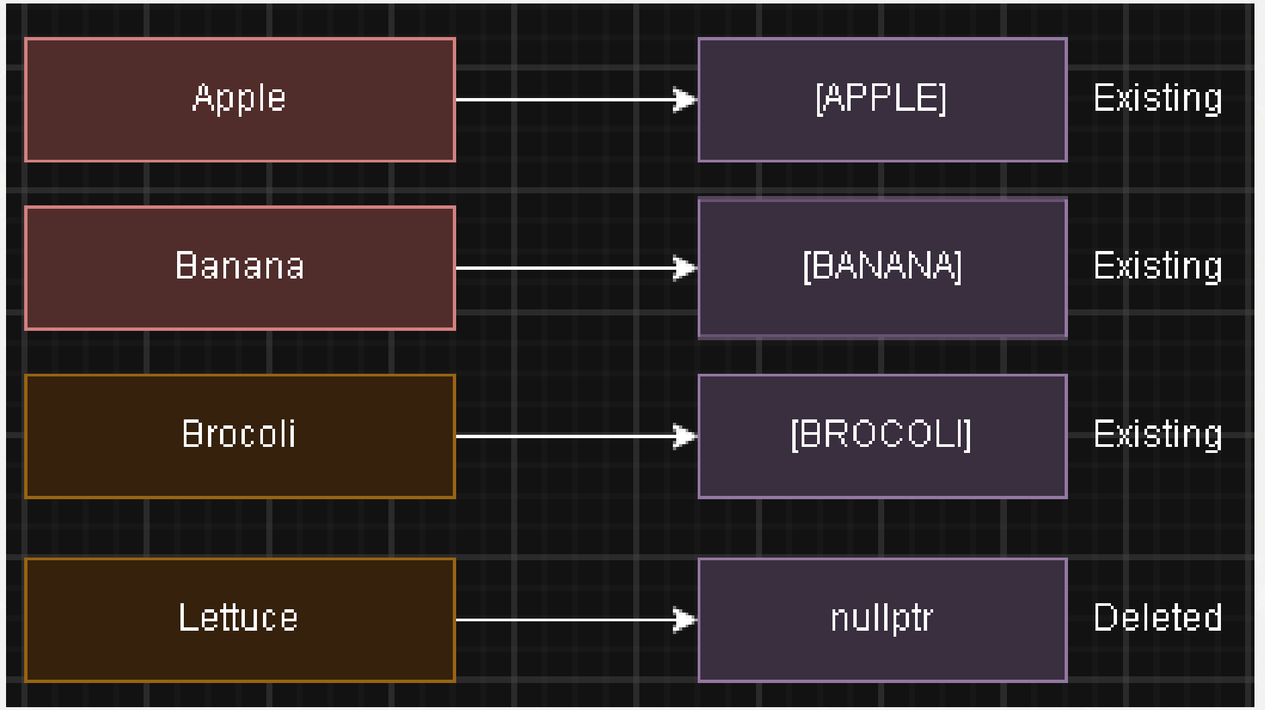
Vegetables:
Vegetables      Broccoli        60          12          720

New Total Amount in Pesos: 870
```

BEFORE

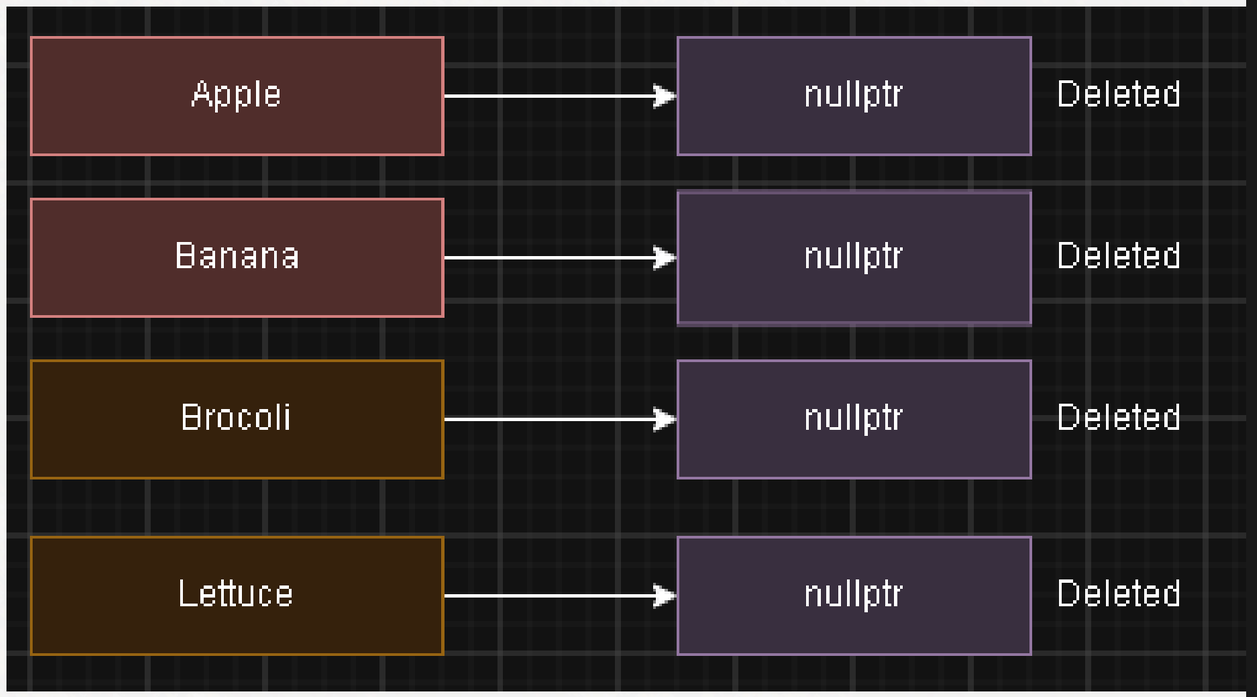
```
// Delete Apple and Banana
    for (int i = 0; i < 2; i++) {
    delete fruitsTaken[i];
    fruitsTaken[i] = nullptr;
}

// Delete Broccoli
    delete vegetablesTaken[0];
```



OUTPUT

AFTER




```
========== POINTER LIST BEFORE FINAL CLEANUP ==========
Apple Pointer    : 0x19b0ea971f0
Banana Pointer   : 0x19b0ea97470
Broccoli Pointer : 0x19b0ea976f0
Lettuce Pointer  : 0
=======================================================

Cleaning up remaining memory...
Fruit Destructor Called: Apple
Fruit Destructor Called: Banana
Vegetable Destructor Called: Broccoli

Memory successfully cleaned up.

========== POINTER LIST AFTER FINAL CLEANUP ==========
Apple Pointer    : 0
Banana Pointer   : 0
Broccoli Pointer : 0
Lettuce Pointer  : 0
=======================================================

All dynamically allocated memory has been deallocated successfully.
No memory leaks remain.
PS C:\Users\mendo\OneDrive\Documents\C++\HoA2Mendoza\output>
```

The End

THANK YOU FOR LISTENING

Nat Mendoza

